Geldium Delinquency Risk: Insights and Recommendation

Report for: Head of Collections, Geldium | Prepared by: Prashant Panchal | Based on the 500-customer dataset (Tasks 1 and 2)

1. Summary of Predictive Insights

Headline: no variable is a strong predictor of delinquency in the current data. Of 500 customers, 80 (16%) are delinquent. Every numeric correlation with delinquency is below 0.06, and a test logistic regression scored AUC ≈ 0.45 (0.50 = chance). The model is not yet ready for live decisions, and the segment gaps below are directional only: none is statistically significant ($p > 0.05$).

Top 3 Risk Factors (candidates to validate)

- **Credit utilization of 30% or more** (Credit_Utilization): 16.8% delinquent vs 12.0% below 30%.
- **Business credit card holders** (Credit_Card_Type): 21.3% vs 14.5% for other card types.
- **Unemployed customers** (Employment_Status): 19.4% vs 15.2% for all others.

Key Insights Summary

Key insight	Customer segment	Influencing variables	Potential impact
Higher utilization goes with higher delinquency (weak)	Customers using 30%+ of their credit limit (417 of 500)	Credit_Utilization	Simple, explainable trigger for early outreach
Some segments run above the 16% average	Business cardholders (108); unemployed (93)	Credit_Card_Type, Employment_Status	Prioritise review; confirm with more data first
Missed_Payments and Credit_Score point the "wrong" way	All customers	Missed_Payments (2.85 delinquent vs 2.99 not); Credit_Score (591 vs 575)	Check target definition and data quality before relying on either field

Note: Month_1 to Month_6 payment history was also inconsistent across months, so it is not ranked as a top factor despite its expected importance.

2. Recommendation Framework

- **Restated Insight:** Customers using 30%+ of their credit limit show a higher delinquency rate (16.8% vs 12.0%). The gap is small and not yet statistically proven, so it should be tested before it drives policy.
- **Proposed Recommendation:** Run a randomised early-outreach pilot for customers at 30%+ utilization, offering support (payment reminders, repayment-plan options), not penalties.
- **Specific:** Contact in-scope customers by SMS/email, with a call for those still above 30% after 30 days. A random half of eligible customers receives nothing, as the comparison group.
- **Measurable:** Cut the 90-day delinquency rate of contacted customers by at least 2 percentage points (about 17% to 15% or below) versus the comparison group.
- **Actionable:** Collections runs outreach with existing channels; Data confirms utilization units and the delinquency definition first; Risk signs off the groups and sample size.
- **Relevant:** Directly supports the goal of reducing delinquency early, using a transparent rule the team can explain.
- **Time-bound:** Month 1: confirm data definitions and sample size. Months 2 to 4: run the pilot. Months 5 to 6: measure results and decide on rollout.
- **Justification and Business Rationale:** Early, supportive contact costs far less than recovering a delinquent account and keeps customers in good standing. The comparison group means the pilot also tests whether utilization really matters, which the current data cannot prove. Stakeholder message: "We will test a simple early-support outreach on customers using 30%+ of their credit, and only expand it if results show it works."

3. Ethical and Responsible AI Considerations

Two fairness risks and mitigations

- **Proxy bias:** Age, Location, Employment_Status and card type can stand in for protected characteristics (for example, customers aged 50 to 64 show a 22.6% rate, and Los Angeles 19.6%). **Mitigation:** keep Age and Location out of the scoring and outreach rules; compare flag rates and error rates across groups (flag-rate ratio of at least 0.80); retest regularly.
- **Unreliable signal on small groups:** segments hold only about 76 to 120 customers each, so a "high-risk" label could unfairly follow a whole group by chance. **Mitigation:** require significance tests and a minimum sample size before any group is flagged; use human review for every action.

Explaining predictions: tell non-technical staff that the score is the estimated chance a customer falls behind, built from a few readable factors (for example, "flagged because 45% of the limit is used"). It is guidance, not a verdict, and the model currently performs near chance.
Responsible use: outreach is supportive, never an automatic denial or penalty; use only the data needed and protect customer privacy; name an owner for monitoring and review; and report model limits openly, including that the current model should not go live until it clearly beats chance.